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Reductive CH_4 Plasma-Induced Modification of Cu Surfaces for Low-Temperature Hybrid Bonding

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***Abstract*—As artificial intelligence (AI) and high-performance computing demand greater parallel data processing, next-generation memory architectures must also evolve. High Bandwidth Memory (HBM) supports this with its 3D-stacked structure and wide I/O interface, but further improvements in power efficiency and thermal management are still required. In addition, microbumps currently used for chip stacking face scalability limitations. To address these challenges, hybrid copper (Cu) bonding has been proposed as a promising solution. This study presents a surface treatment strategy for low-temperature Cu-to-Cu direct bonding using CH_4-based reductive plasma: (1) native Cu oxides were removed, and (2) re-oxidation was suppressed by forming a hydride carbon passivation layer. A design of experiments (DOE) approach was used to optimize ICP (inductively coupled plasma) power, CH_4 flow rate, and pressure. Surface analysis was performed using XPS, TOF-SIMS, TEM, and sheet resistance measurements, while post-bonding evaluation included TEM, SEM, and shear strength analysis. The results demonstrate that the energy of radicals and ions plays a decisive role in Cu oxide reduction, contributing to effective surface activation. The resulting passivation layer was confirmed to be a hydrogenated amorphous carbon (a-C:H) film rich in C, CH, and C_2H species. Bonding was conducted at 260 ℃ and 15 MPa for 1 hour, followed by post-annealing at 200 ℃. TEM analysis revealed a void-free and oxygen-free bonding interface with the presence of carbon, suggesting that the carbon layer acted as a passivating layer. These findings confirm that reductuve CH_4 plasma treatment is highly effective for enabling low-temperature Cu bonding, and highlight the strong potential of this approach for next-generation hybrid bonding technologies.**

***Index Terms*—Cu hybrid bonding, reductive plasma treatment, low temperature bonding, surface modification, carbon passivation**

I. INTRODUCTION

The semiconductor industry made significant advancements at the device level, guided by Moore's Law, from 1975 until around 2012. However, after 2013, progress according to Moore's Law has slowed due to physical limitations, as the channel length approaches the atomic scale. To overcome these limitations, the semiconductor technology roadmap is shifting toward 3-dimensional systems to enhance chip performance and integration density (more than Moore). Currently, Thermo-Compression Bonding (TCB) technology with micro-bumps is being utilized to enable 3-dimensional system integration. From an electrical perspective, including signal and power integrity, it is critical to reduce the pitch size of TCB bumps to sub-micron levels. Moreover, shortening the electrical line length reduces power consumption and alleviates thermal stress. However, reducing the size of TCB bumps below 10 μm presents challenges, especially scaling to sub-micron levels [1]. Here, hybrid copper (Cu) bonding technology has emerged as a key solution, allowing wafer or chip stacking without the need for micro-bumps. Key considerations for hybrid Cu bonding include the antioxidation of the Cu surface, low-temperature bonding, surface activation, Cu dishing control, and minimizing the surface roughness [2, 3]. Among these factors, the removal of Cu oxides is particularly critical, as their presence not only hinders both electrical and mechanical performance but also increases the activation energy required for Cu atom diffusion, which is the main mechanism for Cu hybrid bonding. In addition, preventing re-oxidation of the Cu surface is crucial since Cu readily reacts with atmospheric oxygen to form CuO and Cu_2O [4]. Therefore, an effective plasma treatment is required to remove Cu oxides with minimal surface damage and prevent re-oxidation of the Cu surface in ambient conditions.

There are various plasma treatments, such as Ar, N_2, and O_2, that are used to remove Cu oxide, activate, and protect the Cu surfaces to enhance their performance in bonding applications [5-8]. However, plasma treatments with a high concentration of Ar physically bombard the specimen, which can lead to nonselective etching and potential thinning of the bonding surface, compromising its structural integrity [9]. In contrast, H_2-based plasma treatments, such as forming gas plasma (generally 95% Ar and 5% H_2), provide a chemical pathway for reducing Cu oxide, which can improve the Cu surface for bonding [3, 10]. Conventional H_2 plasma pre-treatment offers higher reduction potential; however, it requires elevated

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substrate temperatures exceeding 150 ℃ to fully reduce native Cu oxide [11]. While forming gas can mitigate surface oxidation to some extent, its low hydrogen concentration (approximately 4–10%) limits its efficacy in achieving complete Cu oxide. Furthermore, when the substrate temperature exceeds 150 ℃, Cu re-crystallization occurs, which promotes the coalescence of fine Cu grains. This phenomenon degrades atomic diffusion and ultimately reduces bonding performance during the Cu bonding process [12]. Ultimately, achieving complete Cu oxide removal using conventional H_2 plasma pre-treatment requires complex processes involving hydrogen concentration and treatment duration, and raises safety concerns and compatibility issues with CMOS mass manufacturing processes.

Therefore, an alternative plasma gas, such as CH_4, is needed. Plasma pre-treatment using Ar-CH_4 is effective and reliable for removing Cu oxide and activating the surface for hybrid Cu bonding, as it leverages both physical bombardment and chemical reactions while generating a significant amount of hydrogen radicals. Hydrogen radicals are highly effective reducing agents. For example, a previous study reported that hydrogen radical treatment removed Cu oxides more effectively than conventional methods such as argon ion beam and FAB etching, achieving complete reduction in just 5 minutes at 443 K, while the latter required at least 13 minutes [13]. This highlights the superior efficiency of hydrogen-based plasma processes. Fig. 1 shows a comparison of various plasma treatments in terms of Cu oxide removal, prevention of re-oxidation, process simplicity, surface stability, and throughput [11, 14-18]. Table 1 compares various plasma treatments on the Cu surface [14, 15, 17, 19].

In this study, the reductive plasma pre-treatment using Ar-CH_4 is investigated for application in Cu hybrid bonding. The key objectives are (1) the removal of Cu oxide from the Cu surface and (2) the prevention of re-oxidation by forming a hydrogenated carbon passivation layer. The reduction of surface Cu oxide using Ar-CH_4 plasma proceeds as follows: First, CH_4 is dissociated into ions, radicals, and electrons. Secondly, these reactive species, particularly the highly reactive hydrogen ions and radicals (H^+, H•), interact with the Cu oxide. When hydrogen radicals encounter Cu oxide or diffuse within it, they react with the oxygen atoms bonded to Cu. This reaction reduces Cu oxide, yielding metallic Cu and water (H_2O). In this process, carbon also plays an important role. It not only participates in reduction reactions but also acts as a passivating agent [20]. Carbon can form a carbonaceous layer that passivates the Cu surface, mitigating the risk of re-oxidation and simultaneously improving bonding properties. This treatment enhances bonding properties, activates the Cu surface, increases its hydrophilicity, improves adhesion, and allows for bonding at lower temperatures, which are critical attributes for advanced 3D stacking applications.

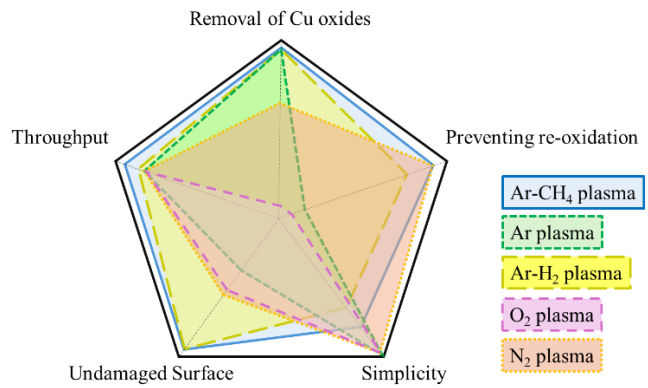

Fig. 1. Performance Mapping of Various Plasma Treatment on Cu Surface

II Experimental Details

A. Plasma Pre-treatment and Bonding Process

The overall experimental procedure is schematically illustrated in Fig 2. An 8-inch notch-type silicon wafer was employed as the starting substrate. A 700 nm thick SiO_2 layer was thermally grown via wet oxidation. Subsequently, a 50 nm titanium (Ti) adhesion layer was deposited to promote adhesion between the substrate and the subsequent metal film. A 200 nm Cu thin film was then deposited using DC magnetron sputtering under a working pressure of 5 mTorr, with 2000 W of DC power and an argon (Ar) gas flow rate of 100 sccm. Subsequently, a 1 μm Cu thin film was deposited by electroplating.

TABLE I
Comparison of Plasma Surface Treatments for Hybrid Cu-to-Cu Bonding.

Plasma gas	Mechanism	Surface Condition After Plasma Treatment	Bonding Temp. (time)	Bonding Pressure	Ref.
Ar	Physical	Dangling bonds	300℃ (60 min)	2 MPa	[17]
Ar/N_2	Physical	Cu_4N and Cu_3N	300℃ (60 min)	0.7 MPa	[14]
O_2	Physical	Amorphous CuO	260℃ (60 min)	16 MPa	[15]
Ar-H_2	Chemical	H residues or cleaned Cu	200℃ (60 min)	5 MPa	[19]
Ar-CH_4	Chemical	Hydrogenated amorphous carbon	260℃ (60 min)	15 MPa	This work

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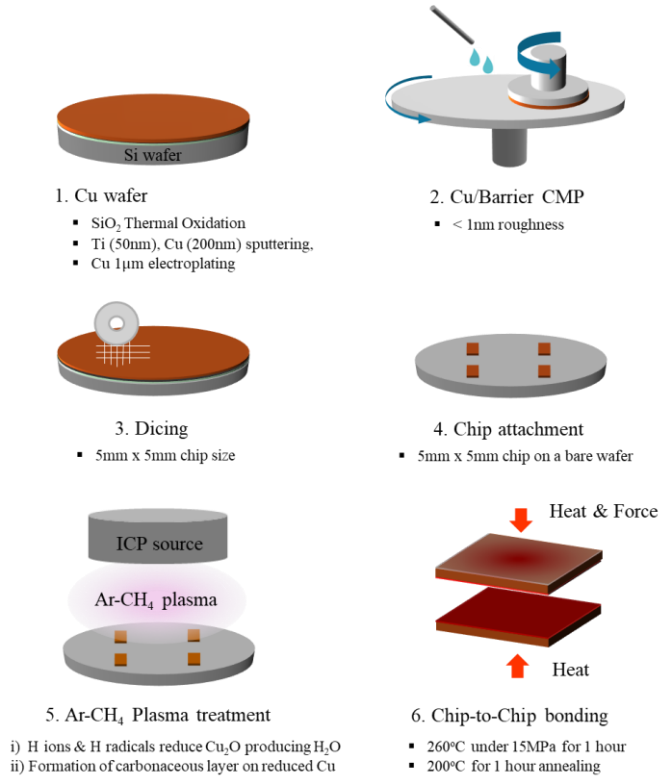

Fig. 2. Schematic representation of the Cu surface pre-treatment and bonding process.

To reduce surface roughness and ensure planarity, Cu chemical mechanical polishing (CMP) was performed on a blanket Cu wafer, resulting in a final surface roughness of less than 1 nm. Photoresist (PR) coating was applied to protect the wafer during mechanical dicing, which was carried out to obtain 5 mm × 5 mm dies. After dicing, the PR layer was removed using acetone and isopropyl alcohol (IPA), and the cleaned samples were temporarily mounted on a bare silicon wafer and subjected to plasma pre-treatment using CH_4 plasma generated by an inductively coupled plasma chemical vapor deposition (ICP-CVD) system (Oxford PP100 Cobra300, S-Fab@SeoulTech) as depicted in Fig. 3.

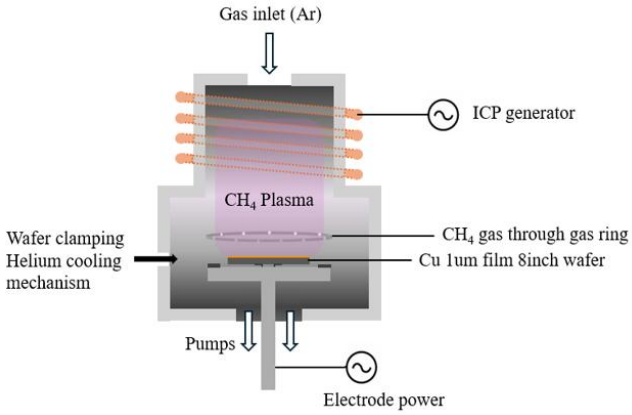

Fig. 3. Inductively Coupled Plasma Chemical Vapor Deposition (ICP CVD).

TABLE II
DOE PARAMETERS FOR
PLASMA PRE-TREATMENT ON CU SURFACE

DOE process parameters			
Case	ICP Power (W)	Pressure (mTorr)	CH_4 Flow rate (sccm)
Case 0	0	0	0
Case 1	1150	6	42.5
Case 2	300	6	42.5
Case 3	1150	10	42.5
Case 4	2000	10	70
Case 5	1150	3	42.5
Case 6	1150	6	42.5
Case 7	2000	6	42.5
Case 8	1150	6	15
Case 9	300	10	70
Case 10	2000	10	15
Case 11	300	3	70
Case 12	1150	6	70
Case 13	300	10	15
Case 14	2000	3	15
Case 15	300	3	15
Case 16	2000	3.68	70

To optimize the plasma treatment conditions, a Design of Experiments (DOE) approach based on Response Surface Methodology (RSM) was employed, in which a Central Composite Design (CCD) was used to systematically vary three key parameters: ICP power, chamber pressure, and CH_4 flow rate, as summarized in Table II. The following parameters were fixed during all experiments: RF power at 0 W; Ar flow rate at 30 sccm; process time of 60 seconds; chuck temperature at 25 ℃; chamber temperature at 60 ℃; and ICP frequency of 13.56 MHz. Among the 16 sample cases, two representative conditions were selected for Cu-to-Cu bonding evaluation. The bonding process was performed under atmospheric conditions at 260 ℃ for 1 hour with an applied pressure of 15 MPa. Post-bonding annealing was conducted at 200 ℃ for 1 hour in air.

B. Surface and Bonding Interface Characterization for Evaluating Cu Bonding Quality

All surface analyses and Cu-to-Cu bonding were performed within one hour after CH_4 plasma treatment, with samples promptly transferred into the analysis chamber or bonding tool to minimize ambient exposure. To assess the effects of plasma treatment on the Cu surface, various surface and interface analysis techniques were employed, including X-ray photoelectron spectroscopy (XPS, Nexsa, Thermo Fisher Scientific Brno s.r.o), time-of-flight secondary ion mass spectrometry (TOF-SIMS, IONTOF GmbH M6), transmission electron microscopy (Cs TEM, NEO ARM/JEOL), and four-point probe measurements.

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XPS measurements were conducted to evaluate the resulting chemical states and composition, with particular focus on quantifying the extent of Cu oxide reduction and characterizing the sp^2/sp^3 hybridization structure of the surface carbon layer. To characterize the microstructure and composition of the C_xH_y layer on Cu, TOF-SIMS and cross-sectional TEM were conducted. In particular, TOF-SIMS was employed to qualitatively assess hydrogen-related species that cannot be reliably analyzed by XPS.

Following bonding, interfacial morphology was examined using scanning electron microscopy (FESEM, JEOL Ltd. (IT-800SHL)) and TEM to identify the presence of micro-voids and to analyze the underlying bonding mechanisms. Bonding quality was evaluated by measuring mechanical shear strength. A shear test was conducted using a multifunction bond tester (MFM 1200, DGFT) to compare the mechanical strength between Ar-CH_4 plasma treated and non-plasma treated samples.

III Results and discussion

The first stage Design of Experiment (DOE) for parameter screening and optimization of Ar-CH_4 plasma conditions was conducted based on X-ray photoelectron spectroscopy (XPS) analysis. Quantification of Cu oxides on the Cu surface was performed through deconvolution of the $Cu2p_{3/2}$ peak, Cu LMM Auger peak, and the analysis of the full width at half maximum (FWHM) of the $Cu2p_{1/2}$ peak. The $Cu2p_{1/2}$ FWHM decreases as the Cu surface becomes more oxygen-rich [21]. Therefore, the $Cu2p_{1/2}$ FWHM was included as a supplementary response parameter to support the evaluation of changes in the Cu surface chemistry under different plasma conditions. To evaluate the surface carbon layer, deconvolution of the C1s peak and measurement of post-etch C-C residue using XPS depth profiling were carried out. The objective of this DOE was to understand the effects of minimizing Cu oxides and simultaneously forming a carbonaceous layer to prevent re-oxidation under ambient conditions. The XPS data used in DOE optimization is shown in Table III. The DOE results are depicted in Fig. 4 with the effect summary which indicates that ICP power and CH_4 flow rate have significant influence on the outcomes.

TABLE III
Response Variables and Optimization Goals in XPS

Objective	Minimization	Maximization
Parameters	▪ Cu_2O intensity in $Cu2p_{3/2}$ ▪ Cu_2O intensity in Cu LMM peak	▪ FWHM of $Cu2p_{1/2}$ ▪ C-C intensity in C1s ▪ Post-etch C-C residue

The optimization priority was set in the following order: oxide reduction, post-etch C-C residue, and C-C intensity. The DOE model was constructed using a second-order polynomial regression (Y) as shown below:

$$Y=\beta_0 + \sum \beta_i X_i + \sum \beta_{ii} X_i^2 + \sum \beta_{ij} X_i X_j + \varepsilon \qquad (1)$$

where Y is the predicted response variable, $\beta_0, \beta_i, \beta_{ii}, and\ \beta_{ij}$ are the regression coefficients, and ε represents the random error. Here, the input variables X_i (I = 1 to 3) correspond to ICP power, chamber pressure, and CH_4 flow rate, respectively. To validate the results from the regression model, Random Forest Model, which is nonlinear Tree-based algorithm, was also employed. DOE results shown in Fig. 4 identifies the two most influential variables for each of the five response parameters and visualizes their effects. Both models consistently indicated that the reduction of Cu oxides was enhanced under conditions of high ICP power, low pressure, and high CH_4 flow rate. These conditions are associated with increased plasma energy and density of reactive species, promoting the generation of reducing agents. Among the 16 DOE conditions, the final selection was the case 16 for its highest oxide reduction efficiency and greatest post-etch carbon residue. The case 11, which showed poor reduction performance but the highest normalized carbon intensity, was also analyzed for comparison with the case 16 in order to investigate the characteristics of the surface carbon layer, as detailed in Table IV.

TABLE IV
XPS results for selected DOE conditions

Case	Non	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cu_2O intensity in $Cu2p_{3/2}$	0.37	0.36	0.52	0.45	0.60	0.57	0.35	0.41	0.38	0.31	0.43	0.42	0.34	0.28	0.38	0.41	0.15
Cu_2O in Cu LMM	0.22	0.24	0.26	0.23	0.24	0.25	0.25	0.22	0.21	0.23	0.31	0.24	0.23	0.25	0.33	0.29	0.17
FWHM of $Cu2p_{1/2}$	1.39	1.37	1.35	1.36	1.32	1.33	1.37	1.44	1.4	1.42	1.32	1.46	1.59	1.37	1.31	1.33	1.52
Post-etch C-C residue	2	3.5	2	2	3.5	4	3	4	1.5	2	3	3	4	2	1	2	7
C-C intensity in C1s	0.21	1.1	7.75	0.34	0.71	0.66	0.69	0.92	0.17	7.92	0.57	17.4	1.68	0.2	0.04	0.18	3.92

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A. Surface Composition of Ar-CH_4 plasma treated Cu surface

X-ray photoelectron spectroscopy analysis was performed using an Al Kα source (1486.6 eV). To investigate the carbon layer and subsurface region of the Cu layer, 5 sequential etching steps of 5 seconds each were applied. As previously mentioned, both case 11 and case 16 were selected for detailed analyses. To compare surface chemical compositions, $Cu2p_{3/2}$ and C1s peak deconvolution results for the non-plasma sample, the case 11, and the case 16 are presented in Fig. 5. The corresponding binding energies of $Cu2p_{3/2}$ and C1s are summarized in Table V [21-30]. For peak deconvolution of $Cu2p_{3/2}$ spectrum, the data obtained after 5 seconds of etching was utilized. As the number of etching steps increases, metallic Cu (Cu^0) peak becomes more prominent in XPS spectra. Therefore, the binding energy of the main Cu peak after etching can be referenced to Cu^0, while the main surface peak prior to etching was positioned between the binding energies of Cu^0 and Cu_2O, assuming a symmetric distribution. The deconvolution of the Cu LMM Auger spectra was performed by assigning the metallic Cu^0 component at 568.67 eV and the Cu^+ (Cu_2O) component at 570.49 eV as seen in Fig. 6 for Case 11 and 16. [22]. The Cu LMM results were incorporated into the DOE not as a standalone oxidation-state identifier, but as an additional response parameter considered alongside the $Cu2p_{3/2}$ analysis to enhance the robustness of the plasma-condition evaluation.

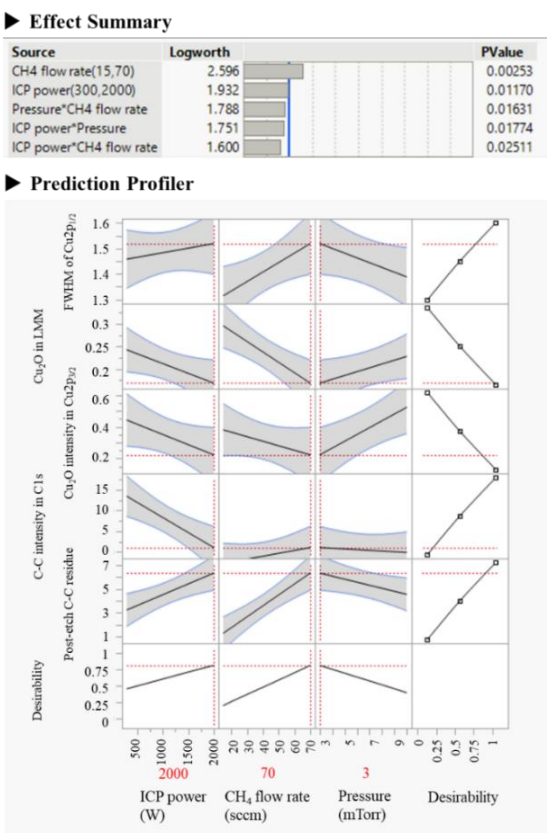

Fig. 4. Effect summary of Process parameters & Prediction Profiler of DOE Model

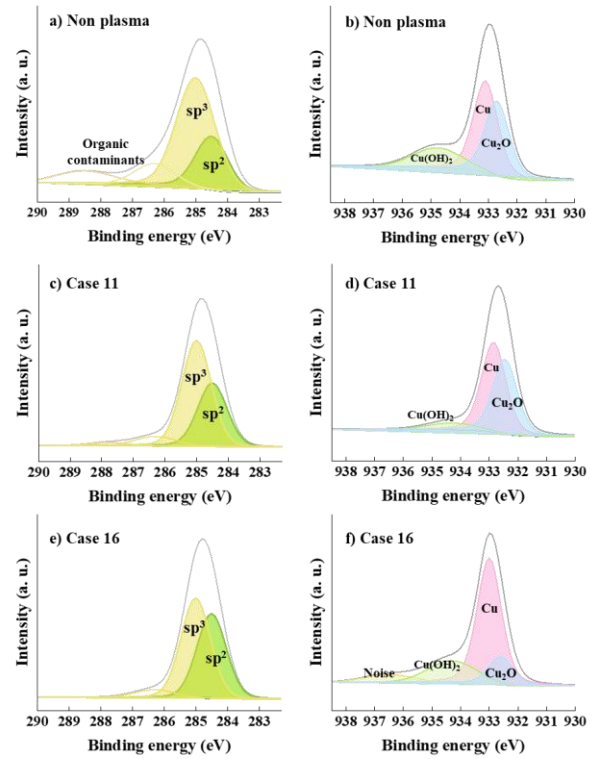

Fig. 5. XPS Deconvolution of $Cu2p_{3/2}$ and C1s for non-plasma, Case 11, and Case 16

TABLE V
BINDING ENERGY OF XPS DECONVOLUTION FOR C1S AND $CU2P_{3/2}$

		C1s			
Peak		C-C (sp^2)	C-C (sp^3)	C-O	C=O
Binding energy (eV)	Non-plasma	284.5	285	286.3	288.5
	Case11	284.5	285	286.3	287.8
	Case 16	284.5	285	286.3	287.8
		$Cu2p_{3/2}$			
Peak		Cu_2O	Cu^0	$Cu(OH)_2$ *The $Cu2p_{1/2}$ FWHM decreases*	
Binding energy (eV)	Non-plasma	932.7	933.1	934.77	
	Case 11	932.85	932.45	934.3	
	Case 16	932.6	933	934.3	

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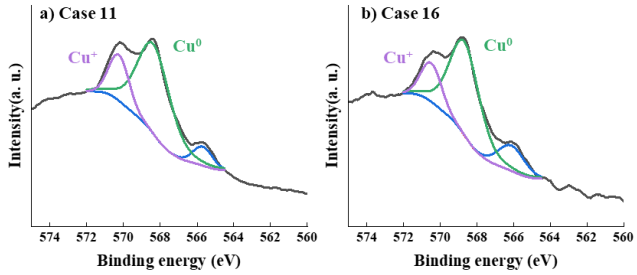

Fig. 6. Cu Auger spectra for (a) Case 11 and (b) Case 16.

In the case 16, a significant decrease in Cu_2O and a corresponding increase in Cu^0 were detected following Ar-CH_4 plasma treatment, indicating that hydrogen radicals and ions generated under high power and low pressure effectively reduced the Cu oxides. In contrast, the case 11 which is processed under the lowest power condition among the DOE cases did not exhibit noticeable oxide reduction. These contrasting results highlight the critical role of hydrogen ions in promoting effective Cu oxide reduction. In addition, an OH peak was also detected in the case 16. However, unlike $Cu(OH)_2$ present in the non-plasma sample, this peak is attributed to surface activation of Ar-CH_4 plasma. This implies the formation of a hydrophilic Cu surface after plasma treatment. On the other hand, the absence of an OH peak in the case 11 suggests that the plasma energy was insufficient to activate the Cu surface.

The C1s spectra indicate that organic surface contaminants were effectively removed after plasma treatment. Notably, the carbonaceous layer formed by Ar-CH_4 plasma treatment contains a high concentration of hydrogen, as confirmed by TOF-SIMS analysis organized in Table VI. Carbon films deposited at low temperatures or with high hydrogen content are generally known to exhibit amorphous structures, as both conditions hinder the formation of long-range of crystalline order. Therefore, the carbonaceous layer formed after plasma treatment is hydrogenated amorphous carbon (a-C:H), consisting of a mixture of sp^2 and sp^3 bonding states. The structural and electrical properties of such films can vary depending on the deposition conditions [31-36].

Accordingly, deconvolution of the C1s spectra was performed to distinguish and quantify the sp^2 and sp^3 components. Compared to the case 11, the case 16 exhibited a higher sp^2 content, which is attributed to the reduced hydrogen incorporation in a-C:H under high ion density and low-pressure conditions [37]. This suggests that the hydrogen content within the carbon film significantly influences its properties. In addition, the mechanical softness of hydrogen-rich a-C:H and the enhanced electrical conductivity of sp^2-rich films suggest that controlling the sp^2/sp^3 ratio through plasma conditions could serve as a key guideline for future bonding recipe optimization.

TABLE VI
Composition of Surface Layer using TOF-SIMS

		Non-Plasma	Case 11	Case 16
Area/Total Area	H-	17.82 %	67.63 %	69.70 %
	C-	1.09 %	2.60 %	2.65 %
	CH-	2.51 %	8.85 %	8.85 %
	CH_2-	0.32 %	1.48 %	1.32 %
	C_2H-	3.52 %	6.47 %	6.18 %
	C_3H-	0.16 %	0.21 %	0.18 %
	C_3H_2-	0.12 %	0.27 %	0.21 %
	C_4H-	0.56 %	0.38 %	0.31 %

B. Evaluation of Surface Roughness and Hydrophilicity

With the RF bias set to zero during CH_4 plasma treatment, ion bombardment was minimized and surface modification was governed primarily by chemical reactions rather than physical sputtering. As a result, the Cu surface roughness was preserved, as confirmed by TEM observations, as shown in Fig. 7, which clearly demonstrate that the Cu surface morphology remained smooth after treatment.

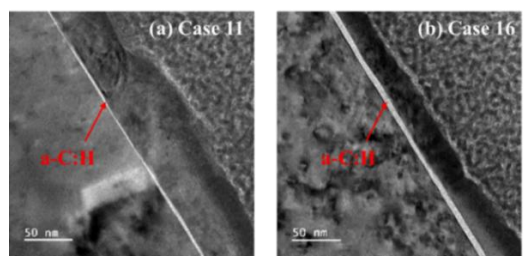

Fig. 7. TEM images of the a-C:H layer after CH_4 plasma treatment: (a) Case 11 and (b) Case 16

XPS O1s deconvolution revealed the presence of OH groups on the a-C:H surface after CH_4 plasma treatment, indicating the incorporation of hydrophilic functional groups, as evidenced by Fig. 8. In addition, TOF-SIMS analysis confirmed an abundant distribution of hydrogen atoms on the treated surface, as presented in Table VI. These observations demonstrate that the a-C:H layer formed by CH_4 plasma possesses good hydrophilicity.

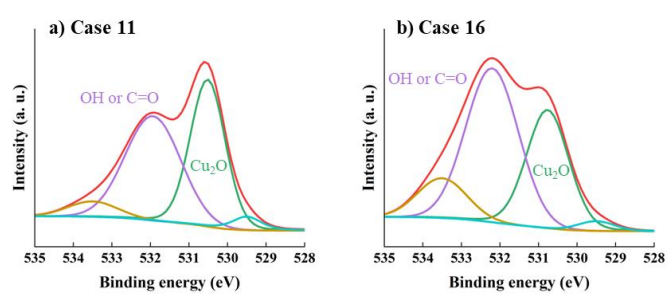

Fig. 8. XPS O1s spectra of a-C:H after CH_4 plasma treatment for (a) Case 11 and (b) Case 16.

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C. Electrical and Microstructural Properties of the Cu surface after plasma treatment

Electrical property of the a-C:H layer deposited on the Cu surface was evaluated through sheet resistance measurement. The sheet resistance was obtained using a four-point probe method, and the resistivity was calculated by multiplying the measured sheet resistance by the thickness of the Cu film. The results are presented in Fig. 9, indicating that plasma-treated samples with a-C:H layer exhibit better conductivity compared to the non-plasma-treated sample, which possesses a native Cu oxide layer. Since the resistivity of thin Cu films is reported to be approximately 2.5 μΩ·cm [37], the measured resistivity of about 2.6 μΩ·cm in the case 16 appears reasonable and sufficiently low. The case 11 exhibited a slightly lower resistivity than the case 16. This may be attributed to the softer nature of the a-C:H layer in the case 11, likely resulting from its higher hydrogen content due to the lower plasma power, which leads to reduced ion energy [36]. The interfacial resistance after bonding may differ from the sheet resistance prior to bonding due to changes in interface chemistry and bonding quality. Electrical characterizations, including Kelvin and daisy chain measurements, after bonding are planned in the following study.

In hybrid bonding structures, CH_4 plasma treatment is simultaneously applied to both the Cu and the dielectric surfaces. CH_4 plasma treatment on O_2-ashed organosilicate dielectrics has been reported to restore carbon content on the dielectric surface and form a dense layer containing C=C bonds, which can reduce leakage current and exhibit high resistivity [38, 39]. This aspect will be further investigated by examining the thickness and structure (particularly the sp^2/sp^3 ratio) of the a-C:H layer.

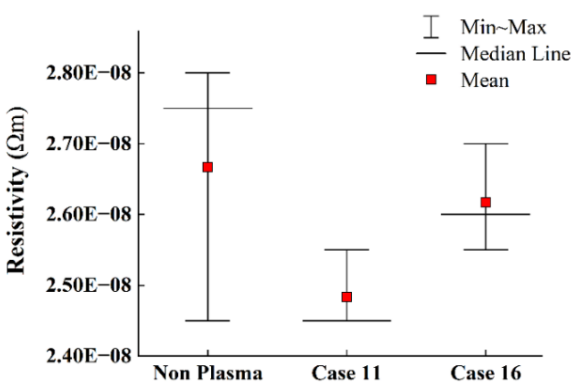

Fig. 9. Resistivity of Cu surface from sheet resistance

The microstructure of the Cu surface before and after Ar-CH_4 plasma treatment was investigated using transmission electron microscopy (TEM) and Fast Fourier Transform (FFT) analysis as shown in Fig. 10. Fig. 10 (a) ~ (f) presents cross-sectional TEM images comparing the surface carbon layers formed in the case 0 (non-plasma), the case 11, and the case 16. A native Cu_2O layer approximately 1~2 nm thick was observed on the Cu surface prior to plasma treatment. In the case 11, a porous a-C:H layer with a thickness of approximately 2 nm was observed. Meanwhile, the case 16 showed a thicker (~ 6.5 nm) and more uniform carbon layer. When oxide reduction is incomplete, the remaining surface oxides interfere with carbon film formation, resulting in poor carbon coverage. Conversely, complete oxide removal leads to a smoother Cu surface that facilitates the deposition of a more continuous and uniform a-C:H layer. Fig. 10 (g) ~ (i) illustrate that in the case 11, where oxide reduction was insufficient, residual Cu_2O was still detected beneath the surface. In contrast, the case 16 exhibited effective oxide removal, with no detectable copper oxides remaining.

D. Bonding interface microstructure and mechanism

After the bonding process, the bonding interfaces were analyzed using cross-sectional TEM and FFT. As shown in Fig. 11, the two selected cases exhibit distinct interfacial characteristics, suggesting different bonding mechanisms. In the case 11, significant Cu diffusion was observed, and a well-defined bonding interface was not discernible. The broadening of the bonding interface can be explained as follows: (1) the interface between Cu and the carbon layer exhibited voids and locally dense carbon clusters, which acted as fast diffusion channels and thereby facilitated significant migration of carbon atoms into the Cu bulk; and (2) incomplete reduction left residual Cu oxides at the interface, disrupting the uniform formation of the a-C:H layer. The coexistence of oxygen and non-uniform a-C:H generated defect sites and mobile C-O species, which further promoted the diffusion of both carbon and oxygen, ultimately leading to a broadened bonding interface.

In contrast, the case 16 clearly demonstrates a carbon-to-carbon bonding interface as shown in Fig. 11. Although a small amount of native oxide remains at the interface between Cu and the carbon layer, no oxides were detected at the bonding interface. This implies that the a-C:H layer formed on the fully reduced Cu surface was uniform and effectively acted as a diffusion barrier preventing re-oxidation of Cu before the bonding process. Additionally, carbon diffusion into the Cu bulk was observed, consistent with the general understanding that Cu-C compounds rarely form. These findings suggest that the diffusion of carbon into the Cu bulk could be strategically utilized in future bonding optimization to achieve a reliable bonding interface. Furthermore, the non-plasma condition showed an average shear strength of 17.0 MPa, while Case 16 exhibited a higher average of 25.0 MPa, indicating improved bonding strength with plasma pretreatment. These average values were obtained from tests conducted on eight specimens for each condition.

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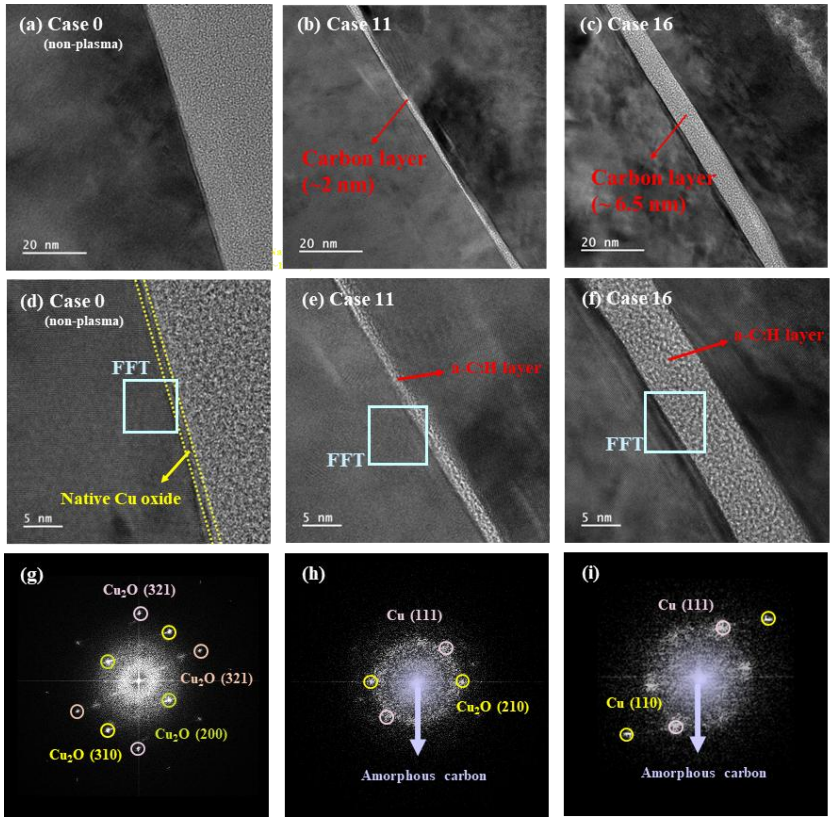

Fig. 10. Surface TEM and FFT Images of Non-Plasma, Case 11, and Case 16: (a) ~(f) show TEM images of each specimen under different magnification, while (g), (h), and (i) present the corresponding FFT patterns for Case 0, Case 11, and Case 16, respectively.

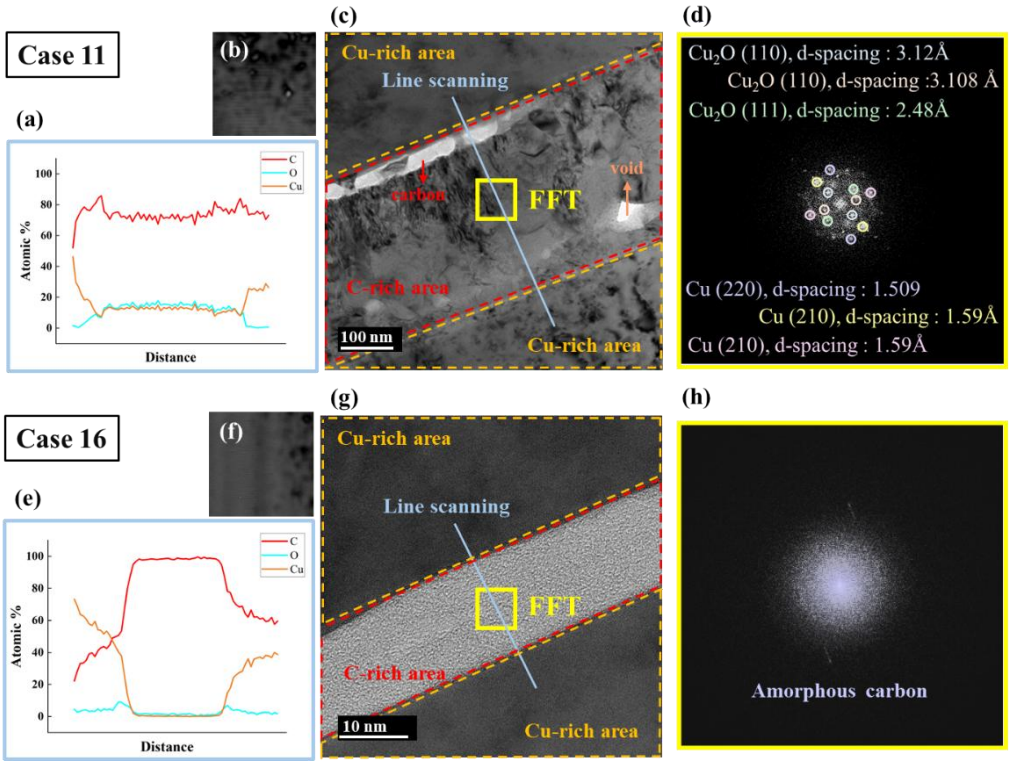

Fig. 11. TEM, SAT, EDS, and FFT analysis of bonding interface microstructures: EDS line profiles of Cu, C, and O (a, e), SAT images (b, f), TEM images (c, g), and FFT patterns (d, h). Figures (a-d) correspond to Case 11, and (e-h) correspond to Case 16.

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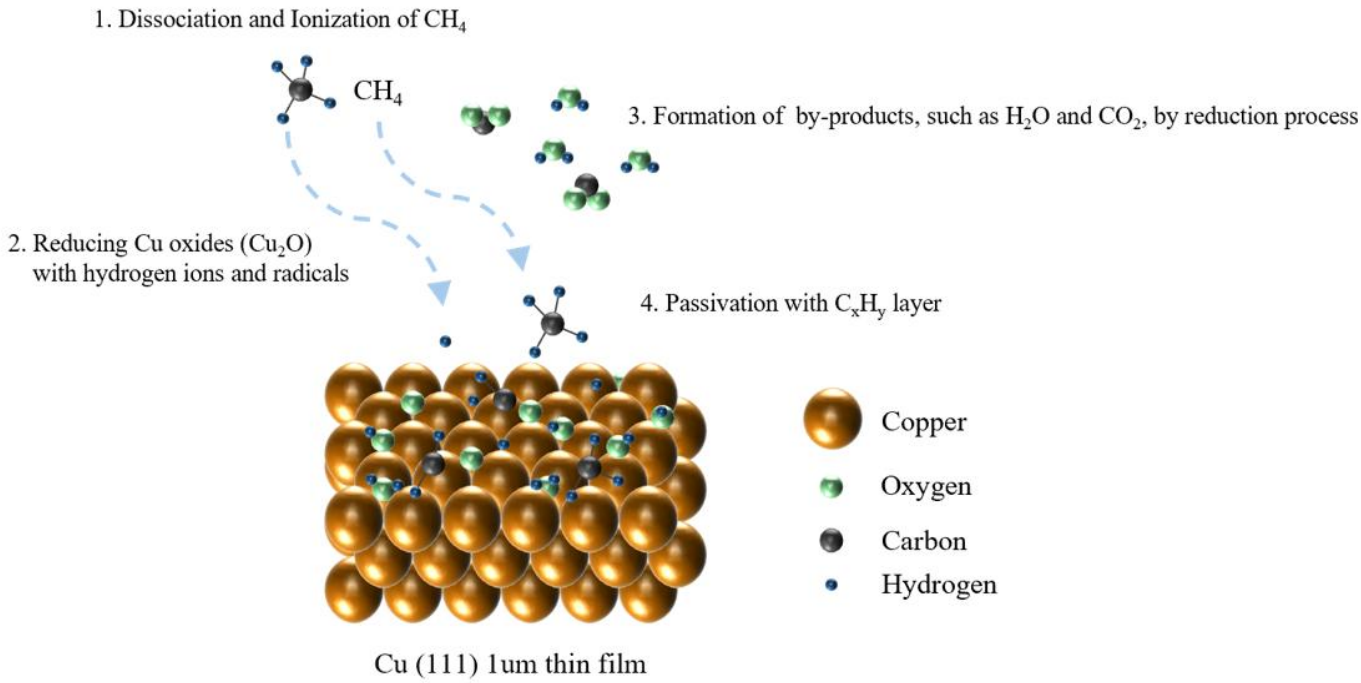

Fig. 12. Schematic Illustration of the Surface Modification Mechanism by Ar-CH_4 Plasma Treatment

E. Mechanism of Ar-CH_4 plasma treatment on the Cu surface

The schematic illustration of the Ar-CH_4 plasma treatment mechanism is presented in Fig. 12, and the reactive species generated in the plasma, such as ions, radicals, and excited molecules, are summarized in Table VII [40, 41]. The concentration and type of these reactive species vary depending on the process parameters.

According to the Ellingham diagram [3], hydrogen ions are thermodynamically more effective reducing agents than hydrogen radicals for Cu oxide reduction. However, the formation of hydrogen ions requires higher energy compared to hydrogen radicals. Therefore, their presence is significantly influenced by plasma power and electron energy levels. An increase in plasma power combined with lower pressure enhances the energy of plasma species, which facilitates ionization and leads to higher production of hydrogen ions. Furthermore, a higher CH_4 flow rate reduces the residence time of gas molecules in the chamber, enabling rapid removal of by-products and improved reduction efficiency.

Based on these principles, the case 16, which was conducted under the highest plasma power and lowest pressure conditions, exhibited the most effective oxide reduction and resulted in a uniform a-C:H layer on the Cu surface. This layer acted as a Cu diffusion barrier [42] that prevented re-oxidation under ambient conditions. In contrast, in case 11, the high CH_4 flow rate and relatively lower plasma power resulted in the premature formation of the a-C:H layer before the complete reduction of surface oxides. As a result, residual oxides remained at the bonding interface, and the a-C:H layer exhibited poor uniformity, as evidenced by the presence of voids and dense carbon clusters in the interfacial region.

TABLE VII
Key Plasma Species Affecting Surface Reactions in Ar-CH_4 plasma

Ions	Radicals	Molecules
H^+, H_2^+, CH^+, CH_2^+, CH_3^+, C_2H^+, $C_2H_2^+$, $C_2H_5^+$, C_3H^+, $C_3H_2^+$, C_4H^+	H, CH, CH_2, CH_3, C_2H_3, C_2H_5	H_2, CH_4, C_2H_2, C_2H_4, C_3H_8

IV Conclusion

This study investigated the Ar-CH_4 plasma treatment on Cu surfaces for hybrid Cu bonding applications, with a focus on surface chemistry. The primary objectives were to effectively reduce native Cu oxides and suppress re-oxidation by forming a carbonaceous layer. A design of experiments (DOE) approach revealed that high plasma power and low pressure enable complete oxide reduction, followed by the formation of a uniform hydrogenated amorphous carbon (a-C:H) layer that acts as a Cu diffusion barrier to prevent re-oxidation. This outcome can be attributed to the enhanced generation of hydrogen ions under such plasma conditions, as high power and low pressure increase electron energy and ionization efficiency, thereby enhancing reduction and promoting the growth of a dense carbon film. Surface analysis using XPS and TOF-SIMS confirmed the presence of hydrogen species within

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the a-C:H layer and showed that the sp^2/sp^3 carbon ratio can be modulated by plasma conditions. Notably, higher plasma power promoted the formation of sp^2-rich C=C bonds, likely due to energetic ion bombardment that suppresses hydrogen incorporation during film growth. These results demonstrate the strong potential of Ar-CH_4 plasma treatment for enabling reliable and oxidation-resistant hybrid Cu bonding. Further optimization of bonding conditions using this plasma approach will be pursued in future work.

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